Chapter 2 — Atomic Structure

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Atomic number is the number of: (a) Nucleons (b) Protons (c) Neutrons (d) Electrons in ion
2. Mass number equals: (a) Protons + neutrons (b) Electrons only (c) Neutrons only (d) Protons + electrons
3. Neutrons in 27Al (Z=13): (a) 14 (b) 27 (c) 40 (d) 13
4. In an electric field, electrons move to: (a) Positive plate (b) Both plates (c) Negative plate (d) No deflection
5. Principal quantum number gives: (a) Subshell shape (b) Shell (c) Spin (d) Orientation
6. Number of p orbitals: (a) 1 (b) 3 (c) 7 (d) 5
7. Maximum electrons in d subshell: (a) 2 (b) 14 (c) 6 (d) 10
8. Hund's rule says: (a) Skip orbitals (b) Fill highest first (c) Singly fill first (d) Pair first

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why does a large jump in successive ionization energies indicate a new shell?
2. Why do electrons deflect more than protons in the same electric field?
3. How do the four quantum numbers describe an electron?
4. Why can an orbital hold only two electrons?
5. Why do electrons occupy degenerate orbitals singly before pairing?
6. How does the Aufbau principle explain 4s filling before 3d?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the four quantum numbers and the shapes of s, p and d orbitals. **(4)**
(b) Describe the rules used to write electronic configurations. **(4)**
Q.4 (a) How do successive ionization energies provide evidence for shells? **(4)**
(b) Describe the behaviour of fundamental particles in an electric field. **(4)**

studysapiens.com